State Space Methods for Flight Vehicles

AE 695

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Systems: Types

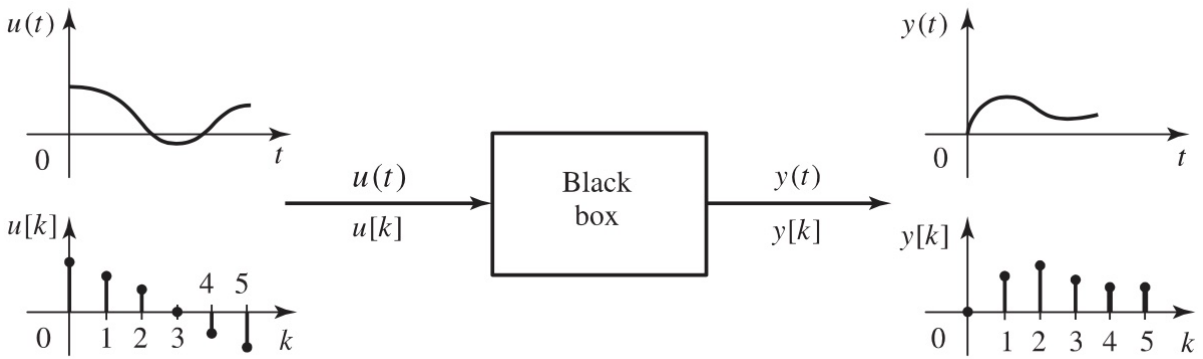
- **Input-Output description**: Transfer function that describes only the terminal properties of the system
- **Internal or state variable description**: A set of differential equations that describes internal as well as terminal behaviors

- **Single-input single-output (SISO) system**: A system with only one input terminal and only one output terminal
- **Multi-input multi-output (MIMO) system**: A system with two or more input terminals and two or more output terminals

- **Continuous-time system**: Continuous-time signals as its input and continuous-time signals as its output
- **Discrete-time system**: Discrete-time signals as its input and discrete-time signals as its output



Systems: Memoryless



Instantaneous or zero memory system

A system is called a memoryless system if its output $y(t_0)$ depends only on input applied at t_0, independent of input applied before or after t_0.

- Most systems have memory; thus, output depends on previous inputs.
- System must be relaxed or at rest to develop the input-output description.
- A system is relaxed if, at time t_1, no energy is stored in the system at that instant. (**Every system is assumed to be relaxed at $t = -\infty$**)
- Under the relaxedness assumption, with operator H, the output can be

$$\boxed{\boldsymbol{y} = H\boldsymbol{u}}$$

Systems: Linearity

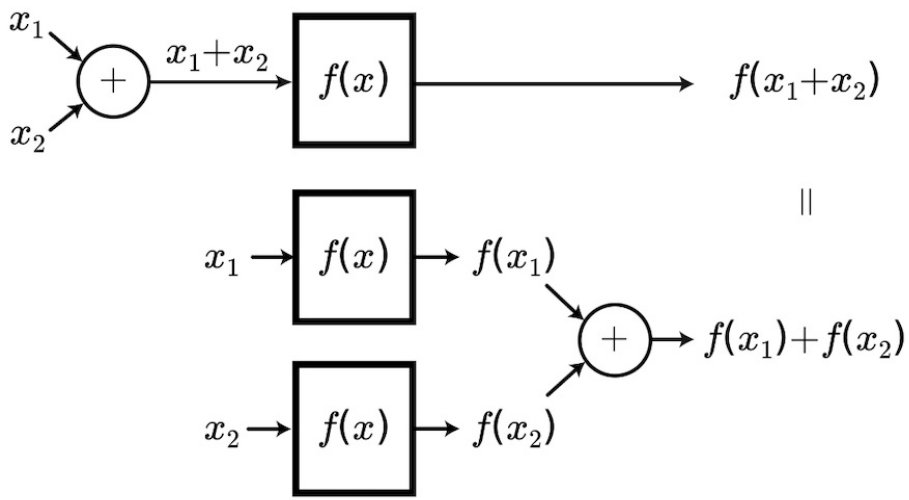
A relaxed system is said to be linear iff

$$H(\alpha_1 \boldsymbol{u}_1 + \alpha_2 \boldsymbol{u}_2) = \alpha_1 H(\boldsymbol{u}_1) + \alpha_2 H(\boldsymbol{u}_2)$$

for any $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$ and for any real numbers α_1, α_2. Otherwise, the relaxed system is nonlinear.

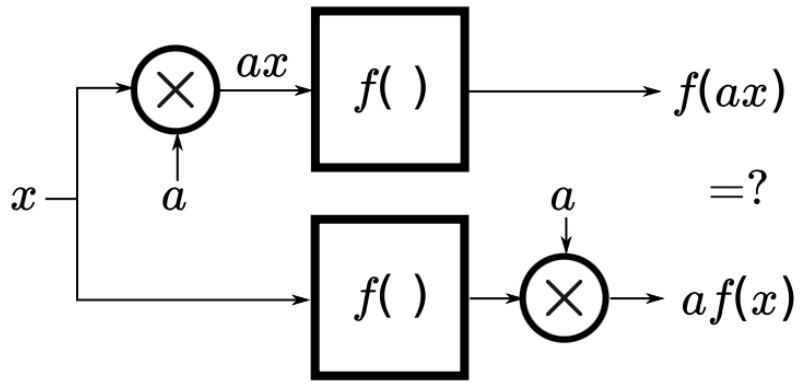
- **Principle of superposition**: Additivity and Homogeneity

$$\boxed{H(\boldsymbol{u}_1 + \boldsymbol{u}_2) = H(\boldsymbol{u}_1) + H(\boldsymbol{u}_2)} \qquad \boxed{H(\alpha \boldsymbol{u}) = \alpha H(\boldsymbol{u})}$$



- **Principle of superposition**: Additivity and Homogeneity

$$H(\boldsymbol{u}_1 + \boldsymbol{u}_2) = H(\boldsymbol{u}_1) + H(\boldsymbol{u}_2) \qquad H(\alpha \boldsymbol{u}) = \alpha H(\boldsymbol{u})$$



- What about the linearity of the system below?

$$y(t) = \begin{cases} \dfrac{u^2(t)}{u(t-1)} & u(t-1) \neq 0 \\ 0 & u(t-1) = 0 \end{cases}$$

- Homogeneity true but not the additivity, thus a nonlinear system.

Example

Consider a system whose input $x(t)$ and output $y(t)$ are related by $y(t) = tx(t)$.

- $x_1(t) \rightarrow y_1(t) = tx_1(t), \quad x_2(t) \rightarrow y_2(t) = tx_2(t)$
- $x_3(t) = ax_1(t) + bx_2(t)$
- $y_3(t) = t(ax_1(t) + bx_2(t)) = atx_1(t) + btx_2(t) = ay_1(t) + by_2(t)$
- Linear system

Example

Consider a system whose input $x(t)$ and output $y(t)$ are related by $y(t) = x^2(t)$.

Nonlinear system



Control Theory: Linearity Test

Example

Check the linearity of the following functions:

- $f(x) = 2x \ \forall \ x \in \mathbb{R}$
- $f(x) = 2x + 9 \ \forall \ x \in \mathbb{R}$
- $f(x) = x^2 + x \ \forall \ x \in \mathbb{R}$
- $f(x) = 3x \ \ \forall \ x \in \{0, 2, 4, 6, \ldots\}$
- $f(x) = 2x$ where $x = \begin{bmatrix} a \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ b \end{bmatrix}, \ a, b \in \mathbb{R}$

- Linear
- Nonlinear (Additivity and Homogeneity)
- Nonlinear (Additivity and Homogeneity)
- Nonlinear (Homogeneity)
- Nonlinear (Additivity)



Linearity

Linear System

A system $\mathcal{L}$ is said to be linear if whenever an input u_1 yields an output $\mathcal{L}(u_1)$ and an input u_2 yields output $\mathcal{L}(u_2)$, we also have

$$\mathcal{L}(c_1 u_1 + c_2 u_2) = c_1 \mathcal{L}(u_1) + c_2 \mathcal{L}(u_2);\ c_1, c_2 \in \mathbb{R}, \mathbb{C}$$

- Any restrictions on the inputs?
- Space of possible inputs should be closed under a linear combination.
- Input $c_1 u_1 + c_2 u_2$ must belong to same space if inputs u_1 and u_2 do so.
- How is this notion of linear systems useful in condensing the input-output description of a system?
- Consider input space with periodic functions having Fourier representation

$$u(t) = \sum_{n=-\infty}^{\infty} \tilde{u}_n e^{j2\pi nt/T},\ \tilde{u}_n = \frac{1}{T}\int_0^T u(t) e^{-j2\pi nt/T} dt$$



Linearity

- Output of the system

$$\mathcal{L}(u(t)) = \mathcal{L}\left(\sum_{n=-\infty}^{\infty} \tilde{u}_n e^{j2\pi nt/T}\right) = \sum_{n=-\infty}^{\infty} \tilde{u}_n \mathcal{L}\left(e^{j2\pi nt/T}\right)$$

- How to specify the system's output for any general input?
- Information about responses to countably infinite set of inputs $e^{j2\pi nt/T},\ n = 0, \pm 1, \cdots$ (sometimes even less)
- Many orders less than all allowable inputs.
- Any other assumptions we made while coming to this conclusion?
- **Extension of additive property to an infinite sum**

$$\mathcal{L}\left(\sum_{1}^{\infty} u_i\right) \neq \sum_{1}^{\infty} \mathcal{L}\left(u_i\right)$$

- **Interchange of operations**: Justified only when system $\mathcal{L}$ is "**sufficiently smooth**".



Issues with Linearity Example

Example

Consider a system defined as follows: The inputs to the system are restricted to being piece-wise continuous functions, bounded over $(-\infty, \infty)$, and having no more than a finite number of discontinuities in a finite time. The system response at any time t is defined as algebraic sum of jumps that are present in the input signal from $-\infty$ up to the current time t.

- **Space of inputs**: Linear space

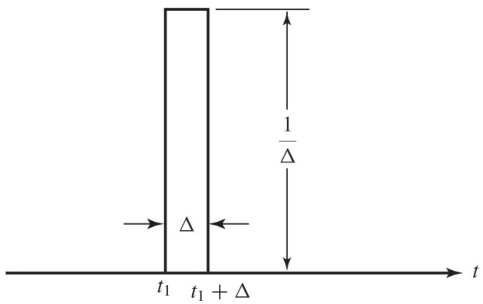
$$\mathcal{L}(c_1 u_1 + c_2 u_2) = c_1 \mathcal{L}(u_1) + c_2 \mathcal{L}(u_2)$$

- Response to a step input to such system is a step output.
- What about its impulse response? Ambiguous !! **Why?**
- Different approximations of impulse lead to different results.
- Response to any continuous approximation of step is also zero.
- Non-smoothness is an issue.



Impulse function or Dirac delta function

- Impulse or Dirac delta function



$$\delta_\Delta(t-t_1) = \begin{cases} 0 & t < t_1 \\ \frac{1}{\Delta} & t_1 \le t < t_1 + \Delta \\ 0 & t \ge t_1 + \Delta \end{cases}$$

$$\delta(t-t_1) \triangleq \lim_{\Delta \to 0} \delta_\Delta(t-t_1)$$

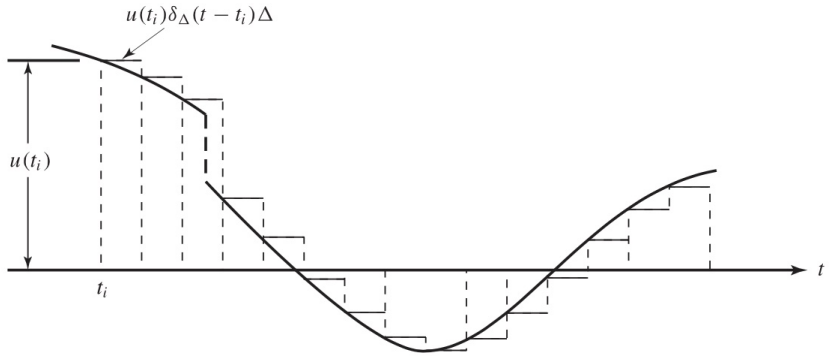
$$\int_{-\infty}^{\infty} \delta(t-t_1)dt = \int_{t_1-\epsilon}^{t_1+\epsilon} \delta(t-t_1)dt = 1$$

- For $f(t)$, continuous function at t_1

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_1)dt = f(t_1)$$

- A relaxed single variable system, with an input-output relation

$$y = Hu$$



- Approximation of input

$$u \approx \sum_i u(t_i)\delta_\Delta(t-t_i)\Delta$$

- For a relaxed linear system

$$y = Hu \approx \sum_i H\delta_\Delta(t-t_i)u(t_i)\Delta$$

Systems: Input-Output Relation

- With a limit $\Delta \to 0$

$$y(t) = \int_{-\infty}^{\infty} [H\delta(t-\tau)]u(\tau)d\tau$$

- What is the meaning of impulse response $g(t,\tau) = H\delta(t-\tau)$?
- Output of relaxed system, due to impulse function input, applied at time τ
- Output at the time instant t

$$y(t) = \int_{-\infty}^{\infty} g(t,\tau)u(\tau)d\tau$$

- A linear relaxed system completely characterized by the impulse response
- Impulse response obtained by direct measurements at input and output
- For system with p input and q output terminals

$$\boldsymbol{y}(t) = \int_{-\infty}^{\infty} \underbrace{\boldsymbol{G}(t,\tau)}_{\text{Impulse response matrix}} \boldsymbol{u}(\tau)d\tau$$

- $g_{ij}(t,\tau)$: Impulse response between j^{th} input and i^{th} output terminal



Causal/Nonanticipatory System

Causal/Nonanticipatory System

A system whose current output depends on past and current inputs but not on future inputs.

- A noncausal system can predict or anticipate future inputs.
- Causality is an intrinsic property of a physical system.
- **Causality**: A necessary condition for a system to be built or implemented in the real world.
- For causal systems,

$$\boldsymbol{y}(t) = H\boldsymbol{u}_{(-\infty,t]} \;\forall\; t \in (-\infty, \infty)$$

- **Noncausal**: Current output depends not only on past but also depends on future input.
- If a system is causal and linear then what about the matrix $\boldsymbol{G}(t,\tau)$?

$$\boldsymbol{G}(t,\tau) = \boldsymbol{0}\,\forall\,\tau \text{ and } \forall\, t < \tau \implies \boxed{\boldsymbol{y}(t) = \int_{-\infty}^{t} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau}$$



Causal System

- Examples of causal system:
  - $y = \int_{-\infty}^{t} x(\tau) d\tau$
  - $y[n] = x[n-1], \quad y[n] = \sum_{k=-\infty}^{n} x[k]$
  - All memoryless systems
- Non-causal systems: $y[n] = x[n] - x[n+1], \quad y(t) = x(t+1)$

Check causality of the systems given by

$$y[n] = x[-n]; \qquad y(t) = x(t)\cos(t+1)$$

- For positive n, $y[n] = x[-n]$ depends on past inputs at past time.
- Not true for negative time, so non-causal system
- What can you say about the second system?
- **Look at the effect of input x only, and not on other functions**
- **Causal system**



Relaxedness of System

A system is said to be relaxed at time t_0 iff the output $\boldsymbol{y}_{[t_0,\infty)}$ is solely and uniquely excited by $\boldsymbol{u}_{[t_0,\infty)}$.

- Relation $\boldsymbol{y} = H\boldsymbol{u}$ applicable only to the relaxed system
- For a system relaxed at time t_0, the input-output relation

$$\boldsymbol{y}_{[t_0,\infty)} = H\boldsymbol{u}_{[t_0,\infty)}$$

- **Necessary and sufficient condition** for the system to be relaxed at t_0

$$\boxed{\boldsymbol{y}(t) = H\boldsymbol{u}_{(-\infty,t_0)} = \boldsymbol{0}\ \forall\ t \geq t_0}$$

- **Net effect of $\boldsymbol{u}_{(-\infty,t_0)}$ on the output after t_0 is zero**.
- **Example**: A time delay system s.t. $y(t) = u(t-1) \forall\ t$
- System is relaxed at t_0 if $u_{[t_0-1,t_0)} \equiv 0$ although $u_{[-\infty,t_0-1)} \neq 0$
- For a relaxed (causal) system, the input output description

$$\boldsymbol{y}(t) = \int_{t_0}^{\infty} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau, \qquad \boxed{\boldsymbol{y}(t) = \int_{t_0}^{t} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau}$$



Relaxedness of System

- Given a system at time t_0, how do we know that the system is relaxed?

A system, described by $\boldsymbol{y}(t) = \int_{-\infty}^{\infty} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau$, is relaxed at t_0 iff

$$\boxed{\boldsymbol{u}_{[t_0,\infty)} \equiv \boldsymbol{0} \implies \boldsymbol{y}_{[t_0,\infty)} \equiv \boldsymbol{0}}$$

- **Necessity**: If system is relaxed at t_0, the output for $t \geq t_0$

$$\boldsymbol{y}(t) = \int_{t_0}^{\infty} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\boldsymbol{\tau})d\tau, \qquad \boxed{\boldsymbol{u}_{[t_0,\infty)} \equiv \boldsymbol{0} \implies \boldsymbol{y}_{[t_0,\infty)} \equiv \boldsymbol{0}}$$

- **Sufficiency**: Need to show that if $\boldsymbol{u}_{[t_0,\infty)} \equiv \boldsymbol{0} \implies \boldsymbol{y}_{[t_0,\infty)} \equiv \boldsymbol{0}$ then system is relaxed at t_0.

$$\boldsymbol{y}(t) = \int_{-\infty}^{\infty} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau = \int_{-\infty}^{t_0} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau + \int_{t_0}^{\infty} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau$$

$$\boxed{\boldsymbol{y}(t) = \int_{-\infty}^{t_0} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau = \boldsymbol{0} \;\forall\; t \geq t_0}$$

- Net effect of $\boldsymbol{u}_{[t_0,\infty)}$ on $\boldsymbol{y}(t)$ for $t \geq t_0$ is zero, thus the system is relaxed.



Relaxedness of System

- Is it practical to check the system output for all time $t \geq t_0$?

If the impulse response matrix $\boldsymbol{G}(t,\tau)$ of a system can be decomposed into $\boldsymbol{G}(t,\tau) = \boldsymbol{M}(t)\boldsymbol{N}(\tau)$, and every element of $\boldsymbol{M}$ is analytic on $(-\infty, \infty)$, then system is relaxed at t_0 iff for some fixed positive ϵ

$$\boldsymbol{u}_{[t_0,t_0+\epsilon)} \equiv \boldsymbol{0} \implies \boldsymbol{y}_{[t_0,t_0+\epsilon)} \equiv \boldsymbol{0}$$

- With $\boldsymbol{u}_{[t_0,\infty)} = \boldsymbol{0}$,

$$\boldsymbol{y}(t) = \int_{-\infty}^{t_0} \boldsymbol{G}(t,\tau)\boldsymbol{u}(\tau)d\tau = \boldsymbol{M}(t) \int_{-\infty}^{t_0} \boldsymbol{N}(\tau)\boldsymbol{u}(\tau)d\tau \;\forall\; t \geq t_0$$

- As $\int_{-\infty}^{t_0} \boldsymbol{N}(\tau)\boldsymbol{u}(\tau)d\tau$ is constant vector, analyticity assumption of $\boldsymbol{M} \implies y(t)$ is analytic on $[t_0, \infty)$.
- If $\boldsymbol{y}_{[t_0,t_0+\epsilon)} \equiv \boldsymbol{0}$ and $\boldsymbol{u}_{[t_0,\infty)} \equiv \boldsymbol{0}$ then $\boldsymbol{y}_{[t_0,\infty)} \equiv \boldsymbol{0}$.
- Relaxedness checked by observing output over **any nonzero interval of time** for the system satisfying the above constraints.



Time Invariance of System

- **Time invariant, fixed, or stationary system**: Characteristics of system do not change with time.
- Shift operator Q_α:

$$\boxed{\bar{u} = Q_\alpha u \iff \bar{u}(t) = u(t-\alpha), \text{ or } \bar{u}(t+\alpha) = u(t)\,\forall\, t}$$

A relaxed system is time-invariant (or stationary or fixed) iff

$$HQ_\alpha \boldsymbol{u} = Q_\alpha H\boldsymbol{u} = Q_\alpha \boldsymbol{y}$$

for any input $\boldsymbol{u}$ and any real number α. Otherwise, the relaxed system is called time-varying.

- For shifted input, the output waveform remains the same except for a shift.
- **Impulse response**: $g(\cdot, \tau) = H\delta(t-\tau)$
- If a linear relaxed system is time-invariant, what condition does that impose on its impulse response?



Time Invariance of System

- For LTI systems,

$$Q_\alpha g(t,\tau) = Q_\alpha H\delta(t-\tau) = HQ_\alpha\delta(t-\tau) = H\delta(t-\tau-\alpha) = g(t,\tau+\alpha)$$

- As shift operator $\boxed{\bar{u} = Q_\alpha u \iff \bar{u}(t+\alpha) = u(t)\,\forall\, t}$

$$\underbrace{g(t,\tau+\alpha)}_{\bar{u}} = Q_\alpha \underbrace{g(t,\tau)}_{u} \implies \boxed{g(t,\tau) = g(t+\alpha,\tau+\alpha)\,\forall\, t,\tau,\alpha}$$

- Using $\tau = -\alpha$ and $\forall\ t,\tau,\alpha$,

$$\boxed{g(t,\tau) = g(t-\tau,0) = g(t-\tau)}$$

- Impulse response $g(t,\tau)$ depends only on the difference $(t-\tau)$
- For the multi-variable case, $\boldsymbol{G}(t,\tau) = \boldsymbol{G}(t-\tau,0) = \boldsymbol{G}(t-\tau)\ \forall\ t,\tau$
- Output of multi-variable LTI and causal system, with $t_0 = 0$

$$\boldsymbol{y}(t) = \int_{t_0}^{t} \boldsymbol{G}(t-\tau)\boldsymbol{u}(\tau)d\tau = \int_{0}^{t} \boldsymbol{G}(t-\tau)\boldsymbol{u}(\tau)d\tau = \int_{0}^{t} \boldsymbol{G}(\tau)\boldsymbol{u}(t-\tau)d\tau$$

- For causal systems, $\boldsymbol{G}(t) = \boldsymbol{0}\,\forall\, t < 0$.



Time-Invariant System

Check the time-invariance property of the system given by

$$y(t) = \sin[x(t)]$$

- Need to check time-invariance properties for **any inputs and any time shift**
- For arbitrary input $x_1(t)$, let output be $y_1(t) = \sin[x_1(t)]$
- Input after shifting by t_0: $x_2(t) = x_1(t - t_0)$
- Output now becomes $y_2(t) = \sin[x_2(t)] = \sin[x_1(t - t_0)]$
- By shifting output y_1 by t_0,

$$y_1(t - t_0) = \sin[x_1(t - t_0)] = y_2(t)$$

- **Invariant system**



Laplace Transform and Transfer Function Matrix

- Using Laplace transform, the output of a system

$$\boldsymbol{y}(s) \triangleq \mathcal{L}(\boldsymbol{y}) = \int_0^\infty \boldsymbol{y}(t)e^{-st}dt, \quad \boldsymbol{y}(t) = \int_0^\infty \boldsymbol{G}(t-\tau)\boldsymbol{u}(\tau)d\tau$$

- Since $\boldsymbol{G}(t-\tau) = \boldsymbol{0} \; \forall \; \tau > t$,

$$\begin{aligned}
\boldsymbol{y}(s) &= \int_0^\infty \int_0^\infty \left(\boldsymbol{G}(t-\tau)\boldsymbol{u}(\tau)d\tau\right) e^{-st}dt \\
&= \int_0^\infty \int_0^\infty \left(\boldsymbol{G}(t-\tau)e^{-s(t-\tau)}d\tau\right) \boldsymbol{u}(\tau)e^{-s\tau}dt \\
&= \int_0^\infty \boldsymbol{G}(v)e^{-sv}dv \int_0^\infty \boldsymbol{u}(\tau)e^{-s\tau}dt \triangleq \boldsymbol{G}(s)\boldsymbol{u}(s)
\end{aligned}$$

- **Transfer function matrix**: $\boldsymbol{G}(s) = \int_0^\infty \boldsymbol{G}(t)e^{-st}dt$
- **Transfer function**: Laplace transform of an impulse response for a single variable
- System is implicitly assumed to be relaxed when transfer function is used.
- **Proper and strictly proper transfer functions/matrices**



Definition of State of System

- Input output description applicable only for initially relaxed systems.
- **Output depends not only on input but also on initial conditions**.

State

The state of the system at time t_0 is the amount of information at t_0 that, together with $\boldsymbol{u}_{[}t_0, \infty)$, determines uniquely the behavior of the system for all $t \geq t_0$.

- **Behavior of system**: All responses including system state
- State at any time **summarizes the essential information** about past inputs required for the **prediction of future behavior**.
- Different inputs might lead to the same state at a given time, but their future behavior would be the same for the same future inputs.



Reference

Reference

- **Chen, Chi-Tsong, "Linear System Theory and Design", CBS College Publishing, 1984.**

